MSG No. 90.96 $P42_121'$ [Type II, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `2.221'`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,6,7,8,9,14,15,16]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[2,3,4,5,10,11,12,13]`

Table 2: Wyckoff site: `2b`, site symmetry: `2.221'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,6,7,8,9,14,15,16]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,3,4,5,10,11,12,13]`

Table 3: Wyckoff site: `2c`, site symmetry: `4..1'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,2,3,6,9,10,11,14]`
2	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[4,5,7,8,12,13,15,16]`

Table 4: Wyckoff site: `4d`, site symmetry: `2..1'`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,6,9,14]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[2,3,10,11]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[4,5,12,13]`
4	$[0,\ 0,\ -z]$	`[7,8,15,16]`

Table 5: Wyckoff site: `4e`, site symmetry: `..21'`

No.	position	mapping
1	$[x,\ x,\ 0]$	`[1,7,9,15]`
2	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ 0\right]$	`[2,5,10,13]`
3	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	`[3,4,11,12]`
4	$[-x,\ -x,\ 0]$	`[6,8,14,16]`

Table 6: Wyckoff site: `4f`, site symmetry: `..21'`

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{2}\right]$	`[1,7,9,15]`
2	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[2,5,10,13]`
3	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[3,4,11,12]`
4	$\left[-x,\ -x,\ \frac{1}{2}\right]$	`[6,8,14,16]`

Table 7: Wyckoff site: 8g, site symmetry: 11'

No.	position	mapping
1	$[x,\, y,\, z]$	[1,9]
2	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z\right]$	[2,10]
3	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z\right]$	[3,11]
4	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	[4,12]
5	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, -z\right]$	[5,13]
6	$[-x,\, -y,\, z]$	[6,14]
7	$[y,\, x,\, -z]$	[7,15]
8	$[-y,\, -x,\, -z]$	[8,16]